

Q12. *Describe the overarching research area of interest. Include full sentences, citations, and persuasion.

What research is currently being done in this research space? Why is this an important area to research? What are gaps in the current research or future directions that have been identified?

You must use at least two (2) primary sources (e.g. peer-reviewed academic journals). How did these sources inform your research and understanding of the topic?

Autonomous robots use sensors like cameras and lidar that work well in ideal conditions, but fail in weather situations such as fog, rain, and low light (Palladin et al., 2025). I am interested in researching how to improve mapping and navigation in urban areas during weather conditions that degrade these sensors. This is an important area to research as there are many urban robots that navigate through urban environments, such as robots for delivery, logistics, and law enforcement. It is necessary to ensure the safety of the environment as they travel in spaces with people.

There is a lot of research working on the problem with many different approaches to the solution. One research is attempting to remove incorrect sensor data caused by environmental noise like rain to create a more accurate map of the environment (Gupta et al., 2025). They compared different algorithms for removing inaccurate data and their ability to do different critical perception tasks. While incorrect data was able to be removed for better obstacle detection, a significant amount of data was removed which localization to fail. This research informs me that when looking at noise reduction in data, it is important to retain as much data as possible. Another research worked on using additional sensors like radar and gated cameras that work better in weather conditions with sensor fusion to improve object proposals during inclement weather (Palladin et al., 2025). Instead of using only lidar for object proposals, the system weights sensor input based on its performance at certain distances and under different weather conditions. They found a significant improvement in low light and weather conditions, although they still saw a drop in performance compared to ideal conditions, especially at longer distances. This research helped me gain a better understanding of object detection.

There are still gaps in the research as there are still issues for autonomous navigation and sensor data during weather situations. The first research mentioned that while it focused on lidar, implementing additional sensors could help with localization and mapping (Gupta et al., 2025). This could potentially be done by combining the denoising algorithms with the multisensory fusion mentioned in the other research. The other research mentioned how most systems rely heavily on lidar and there needs to be more research with additional sensors (Palladin et al., 2025). Gupta, H., Lilienthal, A. J., & Andreasson, H. (2025). Evaluating LiDAR Perception Algorithms for All-Weather Autonomy. *Sensors*, 25(24), 7436. <https://doi.org/10.3390/s25247436>
Palladin, E., Dietze, R., Narayanan, P., Bijelic, M., & Heide, F. (2025). SAMFusion: Sensor-Adaptive Multimodal Fusion for 3D Object Detection in Adverse Weather. *ArXiv.org*. <https://arxiv.org/abs/2508.16408>

Q13. *Briefly describe the approaches that could be taken to investigate this research area of interest. Consider how it could be approached from multiple different disciplines.

When you read the literature, what are some of the anticipated approaches (ie. statistical methods, survey design, computer programming, lab techniques) to answer questions in this research space. Consider how the research area of interest could be approached from multiple different disciplines.

Researching this topic can be taken with multiple approaches, such as computer science, optics, and mathematics. Research on this topic has been conducted by computer science through models for denoising and sensor fusion. Optics are important to understand how signals from sensors are affected by rain and other weather conditions. Research on this topic has also used statistics for data validation from sensor inputs. There are also many mathematical models used for analyzing sensor data and for autonomous navigation. Research in the topic also conducts testing on systems in controlled environments or computer simulated environments requiring computer science. Testing of the system's performance will also need to be compared to ideal conditions and other existing systems, requiring data analysis. Urban robots navigate on sidewalks and roads with pedestrians and other road users. Design of the navigation systems needs to consider accessibility for all road users. A view from the social science and public policy perspective would help in the design constraints of the system to ensure accessibility of all road users.

Q14.

***How will we know that it's a need to be addressed? Are we adhering a problem to a community or population without involving them in the process? How can we avoid this?**

Who does research in this research space? Is anyone being left out?

Who are the stakeholders? What audiences or sectors does this research interest impact? Is anyone being left out?

Many autonomous robots are used in urban environments and college campuses. To ensure the safety of people and objects in the environment, autonomous navigation needs to be improved for low light and weather conditions. Urban robots navigate on roads and sidewalks used by other people, so it is important to understand how people are affected by these systems.

There is a lot of research at universities on autonomous systems, as well as starts like Starship Technologies. Big companies like Waymo from Google and Scout from Amazon are also working on these systems. Communities may be left out in the decisions of these robots which they are tested and operate in. One way that communities can be put in the decision process for this research is by reaching out to hear their thoughts and opinions through surveys. We also must make sure that all community members have access to the surveys or other ways for their thoughts and concerns to be heard.

Autonomous robots are used to benefit the community, through food delivery robots that make food delivery cheaper and law enforcement for surveillance and deterrence. Companies that develop or operate these robots are also stakeholders in this, as they depend on a reliable system for navigation under all conditions.

Q15.

***How did I (we) develop an interest in this topic? Describe why you are well-suited to do research in this area.**

For my high school engineering project, I worked on a system for package logistics in apartments which used autonomous robots to move packages. This led me to looking into robotic systems and autonomous navigation, which I developed an interest in. I am familiar with many navigation algorithms used for this research area. I also have technical experience with c++ and python as well as some experience with Robot Operating System (ROS) used for robotics development. Looking at papers aimed at solving this problem also showed me different approaches to solving the problem, and things that worked and didn't. My experience in robotics and autonomous systems makes me well-suited for this research.

Q16. *Who are the faculty members or researchers currently on campus at UMD who are doing research in this research space. Describe their research and cite at least one (1) additional primary sources (e.g. peer-reviewed academic journals) they have published. How did this source inform your understanding of the topic?

Dr. Pratap Tokekar is an Associate Professor under the Department of Computer Science. He is a member of the Maryland Robotics Center and Center for Machine Learning. He works on robot algorithms that actively sense and gather data environments in noisy and uncertain environments.

In the research Active Asymmetric Multi-Agent Multimodal Learning under Uncertainty, Tokekar and his team worked on addressing problems with data communication between autonomous vehicles. Previous autonomous vehicles that shared data with each other struggled with noisy or corrupted data. Tokekar and his team used a weight-based filtering system that suppressed noise in the data while prioritizing reliable data. They found that this system resulted in improved detection rate of accidents compared to other systems. This informs be about the use of data from other autonomous vehicles, which can help autonomous vehicles see in blind spots. This also taught me that selection of data and processing the data is more important than the amount of data.

Liu, R., Tokekar, P., & Lin, M. (2026). Active Asymmetric Multi-Agent Multimodal Learning under Uncertainty. ArXiv.org. <https://arxiv.org/abs/2602.04763>

Q17. Create an advertisement for the research pitch. (This is not strictly required at this stage. We will share these advertisements in GEMS102 on Thursday Feb. 26. If you do not want your idea to enter the rounds of voting and select "no" for the last question, you do not need to create an advertisement).

Make a video (like a short commercial, about 30-60 seconds in length) or other form of advertisement (PowerPoint, infographic, flyer, etc...)—something that would entice your fellow classmates to be curious about and interested in your topic—this should convince classmates to make your idea into a Gemstone team!

Use your creative ideas here to make this as enticing as possible. This advertisement will be used during various times throughout the semester to quickly and succinctly “plug” your topic/idea for a follow-up. Post the link here.

Be sure that if you use Google Drive or other cloud-based link sharing service that you have your settings set so that anyone with the link can view the advertisement.

N/A

Q18. Do you want this research pitch to be considered for becoming a Gemstone team topic? If you select 'yes' it will be entered into the first round of voting. If you select 'no' your pitch will be used for the purpose of the assignment only and not included in the first vote.

Once you click next the next arrow (>>) after this question, it will submit your Research Pitch form.

- Yes

Embedded Data:

N/A